

Karan Srivastava

Madison, WI | (217)-904-9350 | ksrivastava090198@gmail.com | [LinkedIn://ksrivastava1](https://www.linkedin.com/in/ksrivastava1) | Website: [Karansrivastava.com](https://karansrivastava.com)

EDUCATION

MA, PhD in Mathematics <i>University of Wisconsin-Madison</i> PhD Minor in Computer Science	GPA: 3.7/4	May 2026
BSc. Mathematics <i>University of Illinois at Urbana-Champaign</i> Magna Cum Laude with Highest Distinction in Mathematics	GPA: 3.97/4	May 2020
Study Abroad <i>Independent University of Moscow</i> Graduate Courses in Pure Mathematics, Math in Moscow Program	GPA: 3.87/4	Spring 2019

SKILLS

Languages: Python • Julia • Mathematica • Macaulay2 • Maple • Java • LaTeX • SQL • C++
Libraries / Frameworks: PyTorch • TensorFlow • Keras • Pandas • Scikit-Learn • NumPy • Seaborn • BeautifulSoup
Tools / Platforms: Bash • GitHub([ksrivastava1](https://github.com/ksrivastava1)) • CPLEX • Gurobi • OpenRouter • Docker • Linux computing cluster

EXPERIENCE

IBM, Research Intern	May 2025 - Aug 2025
- Genetically trained a large language model (LLM) to evolve heuristics for vehicle routing, achieving 40% faster convergence on TSP instances vs. standard ALNS, with strong generalization to out-of-distribution problems. - Developed a differential vanishing component analysis framework to discover higher-order dynamical systems from data, outperforming SOTA symbolic regression via numerical linear algebra and optimization.	
IBM, Exploratory Math Sciences Sr Research Intern	May 2024 - Aug 2024
- Developed a novel geometric and optimization framework for discovering scientific formulae from background theory and data with a >98% improvement in speed and 1000x reduction in memory - far exceeding SOTA. - Created a framework for discovering missing polynomial background axioms using computational algebraic geometry with 95% accuracy - a previously unsolved open problem in the automated scientific discovery community. - Designed and released the first benchmark dataset for symbolic regression models incorporating background theory (polynomials and differential equations), now used to evaluate emerging hybrid data-theory approaches.	
University of Wisconsin-Madison and Wisconsin Institute for Discovery, PhD Work	Aug 2020 – Present
- Developed Transformer and Reinforcement Learning models to discover large combinatorial datasets beyond SOTA. - Implemented a Genetic LLM framework to evolve and learn new algorithms for combinatorial problems. - Found bounds for a novel method of reconstructing linear data robust to noise and published findings at the IEEE ISIT 2022. Slides and code can be found on my website.	

SELECTED PUBLICATIONS AND PREPRINTS

- AI Noether - Bridging the Gap Between Scientific Laws Derived by AI Systems and Canonical Knowledge via Abductive Inference with S. Dash et al. 2025. [Arxiv 2509.23004](#)
- SynPAT: A System for Generating Synthetic Physical Theories with Data, with J. Lenchner et al. 2025, [Arxiv 2505.00878](#).
- Generative Modeling for Mathematical Discovery, with J. Ellenberg et al. 2025, [Arxiv 2503.11061](#), Submitted to Advances in Theoretical and Mathematical Physics.
- A Perturbation Bound on the Subspace Estimator from Canonical Projections, with D. Alarcon, [IEEE ISIT 2022](#).

SELECTED PROJECTS

Learning Tic Tac Toe with Reinforcement Learning	July 2023 - Aug 2023
Built a Reinforcement Learning agent for Tic Tac Toe that achieved perfect play in 10,000+ games without heuristics. [Github]	
Erdős Institute Data Science Bootcamp CoverMyMed Copayment Prediction	Nov 2022 - Jan 2023
- Worked in a team of 3 PhDs to <i>train gradient boosting and ensemble models on a 13M-row healthcare dataset</i>. <i>Reduced RMSE from ~\$40 (baseline) to \$15.30.</i> [Github]	
University of Wisconsin Causal Inference through Machine Learning	Aug 2022 - Dec 2022
Designed Support Vector Regression models that can uncover certain causal relationships in synthetic and real-world data with ~90% accuracy on causal detection. [Github] [Blog post on website].	

LEADERSHIP AND SERVICE

Madison Experimental Math Lab , University of Wisconsin-Madison	2022 – Present
Coordinated ~70 undergraduate research projects with faculty and PhD mentors over many semesters.	
Directed Reading Program , University of Wisconsin-Madison	2021 – 2024
Organized a reading program that grouped ~300 undergraduate students with PhD mentors.	
Undergraduate Mentor Program , University of Wisconsin-Madison	2022 – 2024
Founded and organized a mentorship program that paired ~30 PhD mentors with undergraduates.	

CERTIFICATES AND AWARDS

Institute for Foundations of Data Science Research Assistantship - \$25,000 Research Grant	Aug 2023 - May 2024
Campus-wide Exceptional Service Award - Awarded to 3/2300 Teaching Assistants	2022-2023
Exceptional Teaching Award - Demonstrated excellence in teaching for 4 or more semesters	2020-2022